THE MATERNITY METHOD

NCLEX CHEAT SHEET

Newborn Respiratory Complications

Maternal & Newborn › Newborn Complications

The Big Picture At birth, baby's lungs must **clear fluid**, **fill with air**, and stay open, which takes **surfactant**. Preterm babies, C-section babies, and stressed babies can struggle. Know the **signs of distress**, the **4 big conditions**, and the **first steps of resuscitation**.

KNOW THE NUMBERS

> 60	≥ 20 sec	< 100	< 60	3 : 1
Resp rate = tachypnea	Breathing pause = **apnea**	**HR** → start PPV (bag-mask)	**HR** after good PPV → **compressions**	Compressions **to** breaths

MEMORY HACK · RESPIRATORY DISTRESS

Signs baby is working too hard: GRiNCH

G	R	N	C	H
Grunting	**Retractions** (ribs pull in)	**Nasal flaring**	**Central cyanosis**	**High** resp rate (> 60)

The Big Breathing Problems

Condition	Who & why	Signs	Care
Respiratory distress syndrome (RDS)	**Preterm**, infant of **diabetic mom**. Not enough **surfactant** → air sacs collapse	Gets **worse** over hours; grunting, retractions; "ground glass" x-ray	**Surfactant** via ET tube, **CPAP**/ventilator, warmth, prenatal **steroids** prevent it
Transient tachypnea (TTN)	**C-section** without labor, fast labor. **Fluid stays** in the lungs	Fast breathing **soon after birth**; mild	O_2 as needed, support; **clears in 24–72 hrs**
Meconium aspiration (MAS)	**Post-term**, stressed baby passes **meconium** and breathes it in	**Green-stained** skin/fluid, distress, barrel chest	**Vigorous** baby → routine care. **Not vigorous** → PPV; O_2, ventilation; watch for PPHN
Persistent pulmonary HTN (PPHN)	Lung vessels **stay tight** after birth (often with MAS, sepsis, hypoxia)	**Low O_2** despite oxygen; big **pre/post-ductal** O_2 gap	**Quiet**, minimal handling, O_2/ventilation, nitric oxide in NICU
Apnea of prematurity	Immature **breathing center** in the brain	Pauses **≥ 20 sec**, or shorter with **low HR/O_2**	**Gentle stimulation**; **caffeine citrate**; monitor

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Newborn Resuscitation Basics

Goal: **ventilation is the key**

STEP 1
First Steps

- **Warm, dry, stimulate** (rub back, flick soles)
- **Position** the airway ("sniffing" position)
- Suction **only if needed**: **mouth before nose**
- Not breathing well? HR **< 100**?

STEP 2
Ventilate

- **Positive-pressure ventilation (PPV)** with bag-mask
- Watch the **chest rise**; fix the mask/airway if not
- **Pulse ox on the right hand** (pre-ductal)

STEP 3
Compress

- HR **< 60** after 30 sec of effective PPV → **chest compressions**
- **3 compressions : 1 breath**, 100% O_2
- HR still < 60 → **epinephrine** (IV/UVC preferred)

Target O_2 After Birth

Pre-ductal (right hand) SpO_2 **rises slowly**. Don't over-oxygenate:

- **1 min:** 60–65%
- **5 min:** 80–85%
- **10 min:** 85–95%

Too much O_2 in preemies → **retinopathy of prematurity** and lung injury.

Nursing Care

- **Keep warm**: cold stress worsens breathing
- **Position** for open airway; prone/side-lying if allowed
- **Cluster care**; **check glucose**
- Gavage/IV feeds if RR > 60 (aspiration risk)
- Avoid **suctioning** right after surfactant (per protocol)

RED FLAGS · STOP AND CALL THE PROVIDER

Central cyanosis · **Apnea** with bradycardia · **Grunting** that won't stop · **HR < 100** · SpO_2 **below targets** despite O_2

MUST-KNOW FOR THE NCLEX

- **RDS** = not enough surfactant (preterm). **TTN** = leftover fluid (C-section), resolves in 1–3 days.
- **Ventilation** is the most important step in newborn resuscitation.
- Compressions at HR **< 60** after effective PPV; ratio **3:1**.
- Suction **mouth before nose**. Pulse ox on the **right hand**.
- **Caffeine** treats apnea of prematurity.

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QUICK CHECK · TEST YOURSELF

1. A C-section baby breathing 80/min with mild retractions at 2 hrs, improving by day 2, most likely has _____.
2. RDS is caused by a lack of _____.
3. Start chest compressions when the HR is below ____ after effective PPV.
4. Pulse oximetry in resuscitation goes on the _____.
5. The medication used for apnea of prematurity is _____.

Answers: 1. TTN · 2. surfactant · 3. 60 · 4. right hand (pre-ductal) · 5. caffeine citrate

For study use only. Always follow facility protocols, current guidelines, and provider orders.